**AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH Faculty of Engineering**

**Course Name**: Introduction to Electrical Circuits

**Semester:** Spring 2025-26 **Term**: Final **Assignment**:01 **Total Marks**: 20 **Due**:18/5/26

**Question Mapping with Course Outcomes:**

| Item | COs | POIs | K | P | A | Marks | Obtained Marks |
|---|---|---|---|---|---|---|---|
| Q 1-5 | CO4 | | | | | 5 x 4 | |
| | | | | | Total: | 20 | |

**Student Information:**

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**Section:[K]** **Department:CSE**

1. A 150 kVA, 5000V/350V, 100 Hz single-phase transformer has 200 secondary turns, Calculate (a) the primary and secondary currents, (b) the number of primary turns and (c) the maximum value of the flux. **[4]**

2. An 8-pole dc shunt generator with 687 lap-connected armature conductors and running at 380 rpm supplies a load of 11.5 Ω resistance at terminal voltage of 150 V. The armature resistance is 0.36 Ω and the field resistance is 225 Ω.

   Compute the armature current, the induced emf and the flux per pole. **[4]**

3. A stepper motor has a step angle of 3°. Determine (*a*) the resolution, (*b*) the number of steps required for the shaft to make 51 revolutions, and (*c*) the shaft speed, if the stepping frequency is 2700 pps. **[4]**

4. A 3-phase induction motor operates from a supply frequency of 100 Hz. The stator field of the motor rotates at 2500 rpm. Calculate:

   (i) The number of pole of the motor.

   (ii) The speed of the rotor when the slip is 4.5%.

   (iii) The frequency of the rotor when rotor runs at 750 rpm. **[4]**

5. A 3-phase, 16 poles alternator has star-connected winding with 96 turns coil in each phase. The flux per pole is $5\times10^{-2}$ Wb and the speed is 450 rpm. Its stator pitch factor is 0.966 and distribution factor is 0.96. Find (*i*) the frequency, (ii) the rms value of phase voltage and line voltage of this generator, and (*iii*) the three phase voltages as a function of time. **[4]**

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## Ans to the Q NO-01

(a) Given,

$S = 150\ kVA = 150000\ VA$

$V_1 = 5000\ V$, $V_2 = 350\ V$

$f = 100\ Hz$, $N_2 = 200$ turns

Primary and Secondary Currents,

$$I_1 = \frac{S}{V_1} = \frac{150000}{5000}$$

$$= 30\ A$$

$$I_2 = \frac{S}{V_2} = \frac{150000}{350}$$

$$= 428.57\ A$$

(b) Number of Primary Turns,

We know, $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

$$N_1 = \left(\frac{V_1}{V_2}\right) \times N_2$$

$$= \left(\frac{5000}{350}\right) \times 200$$

$$= 2857.14 \text{ turns.}$$

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(C) Maximum Value of Flux;

We knows,

$$E_2 = 4.44 f N_2 \phi_m$$

$$\phi_m = \frac{E_2}{4.44 \times f \times N_2}$$

$$= \frac{350}{4.44 \times 100 \times 200}$$

$$= 0.00394 \text{ Wb}$$

$$= 3.94 \text{ mWb}$$

Ans to the Q NO-02

Given, $P = 8$, $Z = 687$

$N = 380$ rpm

$R_L = 11.5 \Omega$

$V = 150$ V

$R_a = 0.36 \Omega$

$R_{sh} = 225 \Omega$

Lap-connected, $A = P = 8$

We know,

$$I_L = \frac{V}{R_L}$$

$$= \frac{150}{11.5}$$

$$= 13.04 \text{ A}$$

$$I_{sh} = \frac{V}{R_{sh}}$$

$$= \frac{150}{225}$$

$$= 0.67 \text{ A}$$

Armature Current, $I_a = I_L + I_{sh}$

$$= 13.04 + 0.67$$

$$= 13.71 \text{ A}$$

Induced emf, $E = V + I_a R_a$

$$= 150 + (13.71 \times 0.36)$$

$$= 154.94 \text{ V}$$

Flux per pole, $E = \frac{\phi Z N P}{60 A}$

$$\phi = \frac{154.94 \times 60 \times 8}{687 \times 380 \times 8}$$

$$= 0.0356 \text{ Wb}$$

## Ans to the Q NO-03

Given,

Step angle $\beta = 3°$

Revolutions = 51

Frequency $f = 2700$ pps

(a) Resolution, $n = \frac{360°}{\beta}$

$= \frac{360}{3}$

$= 120$ steps

(b) Number of steps for 51 revolutions,

Total steps $= 51 \times 120$

$= 6120$ steps.

(c) Shaft speed, $n = \frac{\beta \times f}{360} \times 60$

$= \frac{3 \times 2700}{360} \times 60$

$= 1350$ rpm.

## Ans to the Q NO-04

Given,

$f = 100\ Hz$

$N_S = 2500\ rpm$

(i) $N_S = \frac{120f}{P}$

$\Rightarrow P = \frac{120 \times 100}{2500}$

$= 4.8$

$\approx 4$ poles (poles are always even)

(ii) $S = 4.5\%$

$= 0.045$

Speed of rotor, $N = N_S(1-S)$

$= 2500 \times (1 - 0.045)$

$= 2387.5\ rpm$

(iii) We know, $S' = \frac{N_S - N}{N_S} = \frac{2500 - 750}{2500}$

$= 0.7$

$f_r = S' \times f = 0.7 \times 100$

$= 70\ Hz$

## Ans to the Q NO-05

Given, $P=16$, $\phi = 5\times10^{-2}$ Wb

$N = 450$ rpm

$T_{ph} = 96$

$k_p = 0.966$

$k_d = 0.96$

(i) Frequency, $f = \frac{PN}{120}$

$= \frac{16\times450}{120}$

$= 60$ Hz

(ii) We know,

$E_{ph} = 4.44 \cdot k_p \cdot k_d \cdot f \cdot \phi \cdot T_{ph}$

$= 4.44 \times 0.966 \times 0.96 \times 60 \times 0.05 \times 96$

$= 1185.83$ V

Line voltage, $E_L = \sqrt{3} \times E_{ph}$

$= \sqrt{3} \times 1185.83$

$= 2053.91$ V

(iii) We know,

$$v(t) = V_m \sin(\omega t \pm \theta)$$

$E_{ph} = 1185.83$

$V_m = \sqrt{2} \times E_{ph}$

$= \sqrt{2} \times 1185.83$

$= 1677.02\ V$

$\omega = 2\pi f = 2 \times 3.1416 \times 60$

$= 377\ rad/s$

We also know that, in 3-phase system between two phase electrical displacement are 120°.

$v_a(t) = V_m \sin(\omega t)$

$= 1677.02 \sin(377t)\ V$

$v_b(t) = V_m \sin(\omega t - 120°)$

$= 1677.02 \sin(377t - 120°)\ V$

$v_c(t) = V_m \sin(\omega t - 240°)$

$= 1677.02 \sin(377t - 240°)\ V$